

REDROB HACKATHON

Refined Screen Specification

S5 Configuration · S7 Results · S9 Analytics · S10 Export · S11 CE Inspector

Every section: exact data shown, exact components, exact interactions. Nothing assumed.

S5 — Ranking Configuration

PURPOSE: Let judges understand and experiment with scoring weights. Every control shown here should have a visible effect on the demo ranking results so judges can verify the system is genuinely parameterized — not a black box.

Layout: Two-Column Split

Left column (60%): all controls. Right column (40%): live preview panel showing how current weight settings would change the top-5 candidates from the last run. If no run yet, right panel shows a placeholder.

Section 1 — Scoring Weights

Six sliders, each 0–100%, always summing to 100%. Show a visual bar below all sliders that fills proportionally as each weight changes.

Control Label	Default Value	What Judges See When Adjusted
Skills Match	35%	Increasing pushes more technically skilled candidates up regardless of career background
Career Quality	30%	Increasing rewards product company + production experience more heavily
Experience Years	10%	Increasing makes YoE the dominant filter — shows the Gaussian curve effect
Location	10%	Increasing moves Pune/Noida candidates up; pushes overseas down noticeably
Education	5%	Increasing shows IIT/NIT graduates rise; confirms education is intentionally low-weighted
Platform Quality	5%	Increasing rewards high GitHub activity + interview completion rate

IMPORTANT: Show the running total at the bottom of the sliders. If the total exceeds 100%, highlight the total in red and disable the Run button until corrected. Weights must always sum to exactly 100%.

Section 2 — Cross-Encoder Settings

Three controls specifically for the CE re-ranking phase:

Control	Type	Default	What It Shows Judges
CE Blend Weight	Slider 0–80%	60%	How much the CE score overrides the algorithmic score. At 0% = pure algorithm. At 80% = CE dominates.

Control	Type	Default	What It Shows Judges
			Moving this live is the most impressive demonstration of the CE phase.
Shortlist Size	Number input 100–300	300	How many candidates go to the CE model. Lower = faster but may miss good candidates near the cutoff. Higher = more thorough but slower.
CE Model	Read-only display	cross-encoder/ms-marco-MiniLM-L-6-v2	Shows the exact model. Not editable — just informs the judge which model is running.

Section 3 — Behavioral Thresholds

Three curve parameter displays. These are NOT hard thresholds — show them as curve descriptions with three sample points each so judges understand the smooth decay:

Signal	Parameter Shown	3 Sample Points to Display
Activity Recency	Half-life: 90 days (editable)	At 30 days inactive → 0.72× At 90 days → 0.37× At 180 days → 0.14×
Recruiter Response Rate	Floor: 0.30 (editable)	At response rate 0.0 → 0.30× At 0.5 → 0.65× At 1.0 → 1.00×
Notice Period	Penalty start: 30 days (editable)	At 30 days → 1.00× At 60 days → 0.69× At 90 days → 0.47×

Section 4 — Execution Mode Toggle

A clearly labeled two-option toggle at the top of the page, visible before any other controls:

Mode	Label	What Changes in the System
Competition Mode (default)	Matches rank.py — No LLM API, cross-encoder only, template-based reasoning	Ensures what judges see is exactly reproducible from the GitHub repo
Demo Mode	Sandbox only — LLM API for richer reasoning if API key is configured	Shows enriched AI-generated reasoning in S8. Clearly marked as Demo Mode throughout.

Section 5 — Action Bar (bottom of page)

- Primary button: 'Run Ranking with These Weights' — triggers S6 pipeline
- Secondary button: 'Reset to Submission Defaults' — restores all values to the competition CSV weights
- Ghost text below buttons: 'Changing weights here does not affect the submitted competition CSV'

S7 — Ranked Results Table

PURPOSE: The primary output screen. Judges spend most time here. Every design decision must serve one goal: let them instantly verify the top 10 are real ML/AI engineers with production experience — not keyword stuffers or honeypots.

Zone 1 — Summary Stats Bar (full width, above table)

A single horizontal row of stats, separated by thin vertical dividers. Shows immediately after ranking completes:

Stat	What It Shows	Example Value
Candidates ranked	How many candidates were in the input	Top 20 of 50 candidates
Runtime	How long the pipeline took	Ranked in 88.2s
Honeypots excluded	Count of candidates given zero score	0 honeypots detected
CE re-ranked	How many candidates went through cross-encoder	CE re-ranked 50 candidates
Avg rank change	Average positions moved due to CE phase	Avg CE rank change ± 6.4

Zone 2 — Filter & Search Bar (below stats, above table)

Six filter controls in one horizontal row. Filtering hides rows but does not re-rank:

Filter	Control Type	What It Filters
Search	Text input	Searches across title, company, location, and reasoning text
Title Category	Dropdown	Core ML/AI Tech Adjacent General Tech Other
Score Range	Two-handle slider	Shows only candidates with score between min and max
Location	Dropdown	Filter by specific city or 'India only' or 'Outside India'
Availability	Toggle	Shows only candidates with open_to_work = true
Reset Filters	Ghost button	Clears all active filters at once

Zone 3 — Results Table (main area)

Full-width table. Sticky header. 20 rows per page. Every row is clickable and opens S8 Candidate Detail drawer.

Column	Width	Exact Data Shown	Visual Detail
Rank	64px	Integer 1–100 inside a circular badge	Ranks 1–10: blue gradient background, white text. Ranks 11–100: neutral background,

Column	Width	Exact Data Shown	Visual Detail
			gray text.
Score	80px	Float to 3 decimal places. Example: 0.987	JetBrains Mono font. Color: green ≥ 0.90 , blue 0.70–0.89, amber 0.50–0.69, gray below 0.50
CE Score	80px	Integer 0–100. Example: 92. Shows 'N/A' if CE unavailable.	Purple pill badge. Purple-50 background, purple-700 text.
Rank Delta	80px	Arrow + number. Example: +12 or -4 or 0	Green badge with up arrow for promoted. Red badge with down arrow for demoted. Gray for stable.
Current Title	160px	Exact current_title from profile. Truncated with ellipsis.	Full title shown on hover tooltip.
YoE	56px	Years to 1 decimal. Example: 7.1 yrs	Green if 5–9 years (ideal range). Gray otherwise.
Location	120px	City name only. Country in hover tooltip.	Plain text. Truncated.
Availability	56px	Colored dot only — no text	Green dot: open + active <30 days. Amber: marginal. Red: unavailable. Hover shows all three signal values.
Reasoning	Remaining	1-2 sentence text, truncated to ~60 characters	Gray body text. Full reasoning shown on hover tooltip. Clicking row opens S8.

Zone 4 — Tier Banding (visual grouping within the table)

Tier	Rows Affected	Visual Treatment
Tier 1	Ranks 1–10 where BOTH score ≥ 0.85 AND CE score ≥ 75	3px green left border on the row + very subtle green-50 row background
Tier 1B	Ranks 1–10 where score ≥ 0.85 BUT CE score <75	3px amber left border. Label: 'Algorithm-Led Pick'. Shows algo and CE disagreed.
Tier 2	Ranks 11–50	Standard white rows. No special treatment.
Tier 3	Ranks 51–100	Very subtle neutral-50 row background to visually differentiate from Tier 2.

Zone 5 — Honeypot Warning Banner (conditional)

Only appears if any top-100 candidate has honeypot_confidence above 0.55. Shows above the table in a red callout: 'Warning: N candidates in this ranking have elevated honeypot confidence. Review their Honeypot tab in Candidate Detail.' In a well-functioning system on clean sample data this banner never appears.

Zone 6 — Pagination (below table)

Shows: 'Showing 1–20 of 100'. Previous and Next buttons. Up to 5 direct page numbers. Page 1 always shows ranks 1–20. This is where judges spend most time.

S9 — Analytics Dashboard

PURPOSE: Answer one question for the judge: does the system actually work as claimed? Every chart here is evidence. The title distribution chart is the most powerful — it proves the system found the 0.75% needle in the 100K haystack.

Zone 1 — Page Header

Title: 'Analytics'. Subtitle: 'Aggregate insights from the last ranking run.' If no run has happened yet: show empty state with message 'Run a ranking first to see analytics.' and a button linking to S4.

Zone 2 — Top 10 Candidate Spotlight (full width, below header)

Ten horizontally scrollable cards, one per top-10 candidate. Each card shows:

Element in Card	Data Shown
Rank badge	Large circular badge: 1 through 10
Title + Company	Current title and company name on two lines
Score	Large score value, color-coded
CE Score	Purple pill badge with CE score
Top 2 matched skills	The two highest-trust skills that matched JD categories
Availability dot	Green/Amber/Red dot
Click action	Opens S8 Candidate Detail for that candidate

Why this exists: NDCG@10 is 50% of the judge score. They will review all 10 top picks carefully. Showing all 10 in one view saves them from opening 10 separate detail drawers.

Zone 3 — Four Chart Row

Four equal-width charts in one horizontal row. Each in a standard card:

Chart	Type	X-axis	Y-axis	Key Insight It Proves
Score Distribution	Histogram	Score buckets: 0.0–0.1, 0.1–0.2, ..., 0.9–1.0	Count of candidates in each bucket	Scores are well-differentiated — not all clustered. Proves scoring discriminates between candidates.
CE vs Algorithm Agreement	Scatter plot	Algorithmic rank (1–300)	Final blended rank (1–100)	Dots far from the diagonal show CE changed the ranking — proves CE adds value beyond the algorithm.
Location	Horizontal bar	Location buckets:	Count of top-100	Confirms location

Chart	Type	X-axis	Y-axis	Key Insight It Proves
Distribution		Pune/Noida, Other Tier-1 India, Other India, Outside India	candidates	scoring worked — India Tier-1 should be over-represented vs full pool.
Availability Breakdown	Donut chart	—	Three segments: Highly Available, Moderate, Low Availability	Confirms behavioral multiplier filtered toward reachable candidates.

Zone 4 — Title Distribution Chart (most important chart on this screen)

A grouped bar chart with two bars per title category — showing the full input pool vs the top-100 ranked output:

Element	Detail
X-axis categories	Core ML/AI titles Tech Adjacent General Tech Low Signal Tech Non-Tech
Bar 1 (left, light)	Percentage of that category in the full input (e.g. Core ML/AI = 0.75% of 100K)
Bar 2 (right, colored)	Percentage of that category in top-100 ranked output (e.g. Core ML/AI = ~65% of top-100)
What judges see	Core ML/AI goes from 0.75% → ~65%. Non-Tech goes from 69% → ~0%. This is the clearest single proof that the system works.
Chart subtitle	'Core ML/AI titles are 87× over-represented in top-100 vs their share of the full pool'

Zone 5 — Disqualifier Impact Table (below charts)

A simple table showing all 6 disqualifier rules and how many candidates in the current ranking were affected by each:

Column	Content
Disqualifier Name	All-consulting career Pure research LangChain only No coding 18m CV/Speech only Closed-source only
Candidates Affected	Integer count. Expected to be low or 0 in small demo samples.
Avg Penalty Applied	The average soft multiplier that was applied: e.g. '×0.41' or 'None'
Note	Even if all values are 0, this table proves the rules exist — which is architecture evidence for Stage 4.

S10 — Export & Submission Panel

PURPOSE: The final step before submission. Must be completely trustworthy — judges need to download a CSV they can submit with zero doubt about its correctness. Every element here is about confidence, not features.

Zone 1 — CSV Preview Table (top of page)

Shows the first 10 rows of the output CSV exactly as they will appear in the downloaded file. Four columns:

Column	Format	Example Value
candidate_id	CAND_XXXXXXX	CAND_0042871
rank	Integer 1–100	1
score	Float 6 decimal places	0.987341
reasoning	1-2 sentence text, no internal newlines	ML Eng, 7.1yrs; shipped FAISS-based semantic search at Meesho. Pune-based, 30-day notice, response rate 0.74.

Below the preview: 'Showing rows 1–10 of 100. Download to see full file.' This sets the right expectation — the preview is not the complete file.

Zone 2 — Validation Results (center of page, most prominent zone)

Runs the same checks as `validate_submission.py` from the hackathon bundle. Shows each check as a row with a green checkmark or red X:

Check Name	Pass Condition	What Failure Means
Row count	Exactly 100 data rows	Submission auto-rejected at Stage 1 — must be exactly 100
Rank completeness	Each integer 1–100 appears exactly once	Missing or duplicate ranks = invalid format
No duplicate candidate IDs	All 100 candidate_ids are unique	Duplicate = format violation
Valid candidate ID format	All IDs match CAND_XXXXXXX pattern	Wrong format = rejected by validator
Scores non-increasing	$\text{score}[\text{rank}=1] \geq \text{score}[\text{rank}=2] \geq \dots \geq \text{score}[\text{rank}=100]$	Out-of-order scores = invalid
Score differentiation	Not all 100 scores are identical	All-same scores signal broken ranker — likely soft rejection
Reasoning non-empty	All 100 reasoning fields contain text	Empty reasoning hurts Stage 4 score
Honeypot rate	Fewer than 10 honeypots in	≥ 10 honeypots = Stage 3

Check Name	Pass Condition	What Failure Means
	top-100	disqualification

Overall result shown in large text below the checklist: 'VALID — Ready to submit' in green, or 'INVALID — Fix errors before submitting' in red. Download button is disabled until all checks pass.

Zone 3 — Download Button

Single large primary button: 'Download submission.csv'. Centered. Only enabled when validation passes. On click, downloads a file named {team_id}.csv as required by the submission spec. Shows a brief success toast: 'Downloaded team_manish.csv (100 rows)'

Zone 4 — Submission Checklist (below download)

Five checklist items. Auto-detects where possible, manual checkbox otherwise:

Item	How Status Is Determined	Action Link
CSV downloaded	Auto: turns green after download button is clicked	—
validate_submission.py passes	Auto: taken from Zone 2 results	—
GitHub repo ready	Manual checkbox + URL input field	Opens GitHub in new tab
Sandbox demo working	Manual checkbox + URL input (pre-filled with current app URL)	Opens sandbox URL
submission_metadata.yaml filled	Manual checkbox	Links to the template file for download

Zone 5 — Metadata Quick Reference (bottom of page, collapsible)

A collapsible section showing the 5 most important fields from submission_metadata_template.yaml that judges check at Stage 3:

Field	Required Value	Where to Set It
reproduce_command	python rank.py --candidates ./candidates.jsonl --out ./submission.csv	submission_metadata.yaml + README
has_network_during_ranking	false — must be exactly false	submission_metadata.yaml
uses_gpu_for_inference	false — must be exactly false	submission_metadata.yaml
pre_computation_required	true — embeddings must be pre-downloaded	submission_metadata.yaml
sandbox_link	Working URL to this demo app	submission_metadata.yaml + portal upload

S11 — Cross-Encoder Inspector

PURPOSE: Prove to methodology judges (Stage 4) that the cross-encoder phase is real, runs correctly, and changes the ranking in meaningful ways. Every module here is evidence of genuine engineering, not a fake wait screen.

Zone 1 — Overview Stats Row (top of page)

Eight metric cards in a horizontal row. These load instantly from the last ranking run:

Metric Card	Exact Value Shown	Why Judges Care
Shortlist Size	N candidates sent to cross-encoder	Confirms how many candidates the CE phase actually evaluated
Pairs Scored	N (same as shortlist — one query per candidate)	Shows the CE ran N forward passes
Model Used	cross-encoder/ms-marco-MiniLM-L-6-v2	Exact model name — judges can verify this is a real, published model
CE Phase Runtime	N.N seconds	Shows the CE took meaningful time — proves it's not a fake pass
Avg CE Score	Mean score 0–100 across all shortlisted candidates	If avg is very high (>80) or very low (<20), CE may not be discriminating well
Algo-CE Correlation	Pearson correlation coefficient, 0–1	0.72 = they partially agree but CE found different signals. 0.99 = CE agreed almost entirely with algo.
Top Promoted	N candidates moved up 10+ positions from CE phase	Key stat — proves CE found candidates the algo underrated
Top Demoted	N candidates moved down 10+ positions from CE phase	Proves CE penalized candidates the algo overrated

Zone 2 — Two-Column Main Area (below stats row)

Left column (50%) contains the Rank Change Scatter Plot. Right column (50%) contains the Biggest Disagreements Table. Both are in standard cards at the same height.

Left: Rank Change Scatter Plot

Element	Detail
X-axis	Algorithmic rank before CE phase (1 to shortlist size, e.g. 1–300). Label: 'Algorithmic Rank'
Y-axis	Final blended rank after CE phase (1–100). Label: 'Final Rank'
Diagonal reference line	Dashed gray line from bottom-left to top-right. Any dot ON this line = CE made no change.
Dot color	Green if CE score ≥80 (CE liked them). Blue if 60–79. Amber if 40–59. Gray if below 40.

Element	Detail
Dot size	All same size. No size encoding needed.
Hover tooltip	Shows: Candidate ID + Title + Algo Rank + Final Rank + CE Score + Rank Delta
Click action	Clicking any dot opens S8 Candidate Detail for that candidate
Key insight for judge	Dots above the diagonal = CE promoted (green dots above = CE found a good candidate the algo missed). Dots below = CE demoted.

Right: Biggest Disagreements Table

Shows the top 10 cases where CE most disagreed with the algorithm — 5 biggest promotions at the top, 5 biggest demotions below. Each row:

Column	Content
Rank change badge	Large +N (green) or -N (red) badge — the visual anchor of the row
Candidate ID + Title	Clickable — opens S8 for that candidate
Algorithmic Rank	Where the algorithm placed them before CE
Final Rank	Where they ended up after CE blending
CE Score	Raw cross-encoder score 0–100
CE Note	One-word reason visible in competition mode: 'Production signals' or 'Keyword mismatch' based on what the passage contained

The most important row in this table: a candidate at algorithmic rank #45 that CE promotes to final rank #7 because their career descriptions showed genuine production deployment work. This is the single most compelling proof that the CE phase adds real value.

Zone 3 — Three Tabs (below the two-column area)

Three tabs with full-width content under each:

Tab 1: CE Score Distribution

Element	Detail
Chart type	Histogram — vertical bars
X-axis	CE score buckets: 0–10, 10–20, 20–30, ..., 90–100
Y-axis	Count of candidates in each bucket
Cutoff line	Vertical dashed blue line showing the CE score of the 100th-ranked candidate — the effective cutoff
Bar colors	Bars above the cutoff line: primary blue. Bars below: neutral gray.
What good looks like	A spread from ~30 to ~95 with a peak around 60–75. A narrow cluster at one value means CE is not discriminating.

Tab 2: Batch Quality Monitor

A table showing one row per batch processed by the cross-encoder. Each batch contains 32 candidates:

Column	Content
Batch #	Sequential number: 1, 2, 3, ... up to N
CE Score Range	Min and max CE score within this batch, e.g. '41 – 88'
Avg CE Score	Mean CE score for the 32 candidates in this batch
Runtime (s)	How long this batch took in seconds
Status	'Success' in green or 'Fallback' in amber — fallback means CE failed and algorithmic score was used

If any row shows Status = Fallback, those candidates used algorithmic score only and their CE Score column in S7 shows 'N/A'. This is handled gracefully — the system still produces a valid CSV.

Tab 3: Full Re-Ranking Log

A complete, searchable, sortable table of all shortlisted candidates and their CE scores. This is the full audit trail:

Column	Content
Candidate ID	Clickable — opens S8
Current Title	From profile
Algorithmic Score	Normalized 0–1 from first-pass scoring
CE Score	Raw 0–100 from cross-encoder
Blended Score	Final score after 40% algo + 60% CE formula
Rank Delta	How many positions they moved: +N or -N or 0
Final Rank	Their position in the final top-100, or 'Not in top-100' if below

Controls above the table: search by candidate ID or title, sort by any column ascending/descending, filter to show only promoted candidates or only demoted. 'Download as CSV' button in the top right exports this entire log for offline analysis.

END OF REFINED SCREEN SPECIFICATION

S5 Configuration · S7 Results · S9 Analytics · S10 Export · S11 CE Inspector
Every zone, every column, every interaction, every edge case.